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United States Patent Application Publication

20250287482

Kind Code

A1

Publication Date

September 11, 2025

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POWER SUPPLY SYSTEM FOR ARC FURNACE AND ASSOCIATED ARC FURNACE AND METHOD

Abstract

Provided is a power supply system for an arc furnace, an arc furnace facility having the same and a method for controlling the power supply system, the power supply system includes a power converter, and a polyphase transformer, a first disconnecting means connecting an input of the power converter to a primary circuit of the transformer, and a second disconnecting means connecting an output of the power converter to the primary circuit of the transformer. The power supply system also includes a first isolation switch and a second isolation switch, and a control circuit.

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Appl. No.: 18/951305

Filed: November 18, 2024

Foreign Application Priority Data

EP	23214193.7	Dec. 05, 2023
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Publication Classification

Int. Cl.: H05B7/148 (20060101); H02J3/00 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to European Patent Application Serial Number EP23214193.7, filed Dec. 5, 2023, which is herein incorporated by reference.

TECHNICAL FIELD

[0002] The present invention concerns arc furnaces, and more particularly a power supply system for arc furnaces, and an arc furnace facility comprising an arc furnace and such a power supply system.

[0003] Arc furnaces are highly nonlinear loads resulting in considerable reactions on the power supply system, in particular so-called flicker.

BACKGROUND

[0004] The document EP 2 329 684 discloses a power supply system to supply a polyphase arc furnace from a grid.

[0005] The power supply system comprises an indirect power converter and a polyphase transformer.

[0006] The indirect power converter supplies a primary circuit of the transformer. A secondary circuit of the transformer supplies electrodes of the arc furnace.

[0007] The indirect power converter is controlled to minimize reactions on the grid in order to suppress the flicker and to comply with grid quality standards.

[0008] During the tuning of the power supply system or when the indirect power converter goes wrong, the power supply system may be faulted out, halted or stopped, for example following reactions on the grid generated by the indirect power converter greater than acceptable reactions.

[0009] During these more or less unplanned and unexpected faults/halts/stops, the arc furnace does not melt scraps to produce molten steel.

[0010] Machines using molten steel, for example rolling mills, are forced to be stopped during stops of the arc furnace, which generates production losses.

[0011] Further, during normal operation of the arc furnace, when the back-to-back electric arc furnace (EAF) converter is in full operation providing (real) power to the EAF furnace while controlling the voltage and current at the EAF, currents delivered by the indirect power converter going through the transformer generates losses and complicates operation of the arc furnace.

[0012] It is therefore proposed a power supply system to overcome wholly or partially the disadvantages related to arc furnace power supply systems known from the prior art.

SUMMARY

[0013] In view of the foregoing, provided is a power supply system for arc furnace including a power converter that includes an input to be connected to a polyphase supply grid and an output, and a polyphase transformer that includes a primary circuit and a secondary circuit, a first disconnecting means, and a second disconnecting means.

[0014] According to an embodiment, the power supply system further includes a first isolation switch and a second isolation switch, the first disconnecting means connecting the input of the power converter to a first end of the first isolation switch, the second disconnecting means connecting the output of the power converter to the first end of the first isolation switch, the second end of the first isolation switch being intended to be connected to at least one electrode of the arc furnace, the second isolation switch connecting the first end of the first isolation switch to the

primary circuit and the secondary circuit being connected to the second end of the first isolation switch, and a third disconnecting means connecting the input of the power converter to the output of the power converter, and a control circuit to control the power converter to supply the electrode and to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, the first isolation switch is open and the second isolation switch is closed or the first isolation switch is closed and the second isolation switch is open, and the third disconnecting means are open, and to control the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second isolation switch is closed, and the third disconnecting means are open or closed.

[0015] Advantageously, the power converter includes an input device, a link circuit comprising a first bus and a second bus, and an output device, wherein: the input device comprises a plurality of input modules each input module being connected to the first bus and the second bus and being intended to be connected to a different phase of the supply grid to supply the link circuit with a continuous voltage from the phase of the supply grid, the output device comprises a plurality of output modules, each output module being connected to the first bus and second bus and to the primary circuit to supply the transformer with an alternative voltage from the circuit link.

[0016] According to an embodiment, each input module including a leg that includes a first string having a first set of input switches connected in series, a second set of input switches connected in series, a first connecting node, a second connecting node, and a third connecting node, the first set of input switches connecting the first bus to the third connecting node, and the first connecting node being disposed between two input switches of the first set of input switches, the second set of input switches connecting the second bus to the third connecting node, and the second connecting node being disposed between two input switches of the second set of input switches, a second string including a first set of input switching units in series, a second set of input switching units in series, and a fourth connecting node, the first set of input switching units connecting the first connecting node to the fourth connecting node, the second set of input switching units connecting the second connecting node to the fourth connecting node, and the fourth connecting node being intended to be connected to the phase of the supply grid, the control circuit being configured to control the input switches, the input switching units, and the output modules, the third connecting nodes of the input modules being connected together.

[0017] Advantageously, each input module including a leg that includes a first string having a first input switch, a second input switch, a first connecting node, a second connecting node, a third connecting node, a first set of input switching units connected in series, and a second set of input switching units connected in series, the first input switch connecting the first bus to the first connecting node, and the first set of input switching units connecting the first connecting node to the third connecting node, the second input switch connecting the second bus to the second connecting node, and the second set of input switching units connecting the second connecting node to the third connecting node, a second string including a third input switch, a fourth input switch, and a fourth connecting node, the third input switch connecting the first connecting node to the fourth connecting node, and the fourth input switch connecting the second connecting node to the fourth connecting node, the fourth connecting node being intended to be connected to the phase of the supply grid, the control circuit being configured to control the input switches, the input switching units, and the output modules, the third connecting nodes of the input modules being connected together.

[0018] According to an embodiment, each input module including a leg that includes a first string having a first input switch, a second input switch, a first connecting node, a second connecting node, a third connecting node, a first set of input switching units connected in series, and a second set of input switching units connected in series, the first set of input switching units connecting the

first bus to the first connecting node, and the first input switch connecting the first connecting node to the third connecting node, the second set of input switching units connecting the second bus to the second connecting node, and the second input switch connecting the second connecting node to the third connecting node, a second string including a third input switch, a fourth input switch, and a fourth connecting node, the third input switch connecting the first connecting node to the fourth connecting node, and the fourth input switch connecting the second connecting node to the fourth connecting node, the fourth connecting node being intended to be connected to the phase of the supply grid, the control circuit being configured to control the input switches, the input switching units, and the output modules, the third connecting nodes of the input modules being connected together.

[0019] Advantageously, each input module including a string that includes a connecting node being intended to be connected to the phase of the supply grid, a first set of input switching units connected in series, and a second set of input switching units connected in series, the first set of input switching units connecting the first bus to the connecting node, and the second set of input switching units connecting the second bus to the connecting node, and the control circuit being configured to control the input switching units, and the output modules.

[0020] According to an embodiment, each output module including a leg that includes a first string having a first set of output switches connected in series, a second set of output switches connected in series, a first connecting node, a second connecting node, and a third connecting node, the first set of output switches connecting the first bus to the third connecting node, and the first connecting node being disposed between two output switches of the first set of output switches, the second set of output switches connecting the second bus to the third connecting node, and the second connecting node being disposed between two output switches of the second set of output switches, a second string including a first set of output switching units in series, a second set of output switching units in series, and a fourth connecting node, the first set of output switching units connecting the first connecting node to the third connecting node, the second set of output switching units connecting the second connecting node to the fourth connecting node, and the fourth connecting node being intended to be connected to a phase of the arc furnace, the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

[0021] Advantageously, each output module including a leg that includes a first string having a first output switch, a second output switch, a first connecting node, a second connecting node, a third connecting node, a first set of output switching units connected in series, and a second set of output switching units connected in series, the first output switch connecting the first bus to the first connecting node, and the first set of output switching units connecting the first connecting node to the third connecting node, the second output switch connecting the second bus to the second connecting node, and the second set of output switching units connecting the second connecting node to the third connecting node, a second string including a third output switch, a fourth output switch, and a fourth connecting node, the third output switch connecting the first connecting node to the fourth connecting node, and the fourth output switch connecting the second connecting node to the fourth connecting node, the fourth connecting node to be connected to a phase of the arc furnace, the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

[0022] According to an embodiment, each output module including a leg that includes a first string having a first output switch, a second output switch, a first connecting node, a second connecting node, a third connecting node, a first set of output switching units connected in series, and a second set of output switching units connected in series, the first set of output switching units connecting the first bus to the first connecting node, and the first output switch connecting the first connecting node to the third connecting node, the second set of output switching units connecting the second

bus to the second connecting node, and the second output switch connecting the second connecting node to the third connecting node, a second string including a third output switch, a fourth output switch, and a fourth connecting node, the third output switch connecting the first connecting node to the fourth connecting node, and the fourth output switch connecting the second connecting node to the fourth connecting node, the fourth connecting node to be connected to a phase of the arc furnace, the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

[0023] Advantageously, each output module comprises a string including a connecting node being intended to be connected to a phase of the arc furnace, a first set of output switching units connected in series, and a second set of output switching units connected in series, the first set of output switching units connecting the first bus to the connecting node, and the second set of output switching units connecting the second bus to the connecting node, the control circuit being configured to control the output switching units and the output modules.

[0024] According to an embodiment, the power supply system further includes a third isolation switch, the third isolation switch connecting the secondary circuit to the second end of the first isolation switch, the control circuit being configured: to control the power converter to supply the electrode and to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, the first isolation switch is open and the second and third isolation switches are closed or the first isolation switch is closed and the second and third isolation switches are open, and the third disconnecting means are open, and to control the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second and third isolation switches are closed, and the third disconnecting means are open or closed.

[0025] Another embodiment of the invention relates to an arc furnace facility.

[0026] The arc furnace facility includes an arc furnace including at least one electrode and a power supply system as defined above, wherein the electrode is connected to at least one phase of the secondary circuit of the polyphase transformer.

[0027] Advantageously, the arc furnace includes a plurality of electrodes, each electrode being connected to a different phase of the secondary circuit.

[0028] According to an embodiment, the power supply system further includes a rectifier connected to the electrode and to each phase of the secondary circuit to supply the electrode with a continuous voltage from the secondary circuit.

[0029] Another embodiment of the invention relates to a method for controlling a power supply system of an arc furnace, the power supply system including a power converter comprising an input connected to a polyphase supply grid and an output, and a polyphase transformer including a primary circuit and a secondary circuit, first disconnecting means, and second disconnecting means.

[0030] The method includes controlling the power converter to supply the electrode and to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, a first isolation switch is open and a second isolation switch is closed or the first isolation switch is closed and the second isolation switch is open, and third disconnecting means are open, and controlling the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second isolation switch is closed, and the third disconnecting means are open or closed, the first disconnecting means connecting the input of the power converter to a first end of the first isolation switch, the second disconnecting means connecting the output of the power converter to the first end of the first isolation switch, the second end of the first

isolation switch being intended to be connected to at least one electrode of the arc furnace, the second isolation switch connecting the first end of the first isolation switch to the primary circuit and the secondary circuit being connected to the second end of the first isolation switch, and the third disconnecting means connecting the input of the power converter to the output of the power converter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] Other characteristics and advantages of the invention will emerge on reading the following description of embodiments of the invention, provided solely by way of non-limiting examples and with reference to the drawings in which:

[0032] FIG. 1 illustrates schematically a first embodiment of an arc furnace facility according to the invention,

[0033] FIG. 2 illustrates schematically an embodiment of a power converter according to the invention,

[0034] FIG. 3 illustrates schematically a first embodiment of an input module according to the invention,

[0035] FIG. 4 illustrates schematically a second embodiment of the input module according to the invention,

[0036] FIG. 5 illustrates schematically a third embodiment of the input module according to the invention,

[0037] FIG. 6 illustrates schematically a fourth embodiment of the input module known from the prior art,

[0038] FIG. 7 illustrates schematically a first embodiment of an output module according to the invention,

[0039] FIG. 8 illustrates schematically a second embodiment of the output module according to the invention,

[0040] FIG. 9 illustrates schematically a third embodiment of the output module according to the invention,

[0041] FIG. 10 illustrates schematically a fourth embodiment of the output module

[0042] known from the prior art,

[0043] FIG. 11 illustrates schematically a first embodiment of input switches and output switches according to the invention,

[0044] FIG. 12 illustrates schematically a second embodiment of the input switches and the output switches according to the invention,

[0045] FIG. 13 illustrates schematically a third embodiment of the input switches and the output switches according to the invention,

[0046] FIG. 14 illustrates schematically a first embodiment of input switching units and output switching units according to the invention,

[0047] FIG. 15 illustrates schematically a second embodiment of the input switching units and the output switching units according to the invention, and

[0048] FIG. 16 illustrates a second embodiment of the arc furnace facility according to the invention.

DETAILED DESCRIPTION

[0049] FIG. 1 illustrates a first embodiment of an arc furnace facility.

[0050] The arc furnace facility comprises a polyphase power supply system 1 connected to a grid G and an arc furnace 2 connected to the power supply system 1.

[0051] The power supply system 1 supplies the arc furnace 2 from electrical energy supplied by the

grid **G**.

[0052] It is assumed that the power supply system **1** and the polyphase arc furnace **2** have each three phases, the grid **G** being a three phases supply grid.

[0053] Each of the power supply system **1** and the polyphase arc furnace **2** may have two phases or more than three phases, the grid **G** having the same number of phases as the power supply system **1**.

[0054] The arc furnace **2** comprises a set of electrodes comprising three electrodes **2a**, **2b**, **2c**.

[0055] The electrodes **2a**, **2b**, **2c** supplied by the power supply system **1** generate electric arcs between the electrodes **2a**, **2b**, **2c** to melt the contents of the arc furnace **2**, for example scrap steel.

[0056] The power supply system **1** comprises three inputs **3**, **4**, **5**, each input being connected to a different phase of the grid **G**, and three outputs **6**, **7**, **8**.

[0057] Each output **6**, **7**, **8** is connected to an electrode **2a**, **2b**, **2c**.

[0058] The power supply system **1** further comprises a power converter **9**, a transformer **10**, a control circuit **11**, current sensors, and voltage sensors.

[0059] The control circuit **11** may comprise a processing unit.

[0060] The power supply system **1** further comprises first disconnecting means **13**, second disconnecting means **14** and third disconnecting means **270**.

[0061] Each disconnecting means of the first, second and third disconnecting means **13**, **14**, **270** comprises for example a disconnecter.

[0062] The power supply system **1** comprises a first isolation switch **15A**, a second isolation switch **15B** and a third isolation switch **15C**.

[0063] The first, second and third disconnecting means, and the first, second and third isolation switches **15A**, **15B**, **15C** are controlled by the control circuit **11**.

[0064] The power converter **9** comprises three inputs **16**, **17**, **18** and three outputs **19**, **20**, **21**.

[0065] Each input **16**, **17**, **18** of the power converter **9** is connected to a different input **3**, **4**, **5** of the power supply system **1**.

[0066] The transformer **10** further comprises a primary circuit including three phases **22**, **23**, **24** and a secondary circuit including three phases **25**, **26**, **27**.

[0067] Each phase **25**, **26**, **27** of the secondary circuit is connected to a different output **6**, **7**, **8** of the power supply system **1**.

[0068] Each phase **25**, **26**, **27** of the secondary circuit comprises a sensing module **300**, **301**, **302** comprising a voltage sensor and a current sensor measuring a voltage and a current of the phase.

[0069] The third disconnecting means **270** connect the inputs **16**, **17**, **18** of the power converter **9** to the outputs **19**, **20**, **21** of the power converter **9**.

[0070] The first disconnecting means **13** connect the input **16**, **17**, **18** of the power converter **9** to a first end of the first isolation switch **15A**.

[0071] The second disconnecting means **14** connect the outputs **19**, **20**, **21** of the power converter **9** to the first end of the first isolation switch **15A**.

[0072] The second end of the first isolation switch **15A** is connected to the outputs **6**, **7**, **8** of the power supply system **1**.

[0073] The second isolation switch **15B** connects the first end of the first isolation switch **15A** to the phases **22**, **23**, **24** of the primary circuit and the third isolation switch **15C** connects the second end of the first isolation switch **15A** to the phases **25**, **26**, **27** of the secondary circuit.

[0074] The first, second and third isolation switch **15A**, **15B**, **15C** are interlocked so that when the first isolation switch **15A** is open, the second and third isolation switches **15B**, **15C** are closed, and when the first isolation switch **15A** is closed, the second and third isolation switches **15B**, **15C** are open.

[0075] In another embodiment, the power supply system **1** does not comprise the third isolation switch **15C**, the second end of the first isolation switch **15A** being connected to the phases **25**, **26**, **27** of the secondary circuit.

[0076] FIG. 2 illustrates an embodiment of the power converter 9.

[0077] The power converter 9 further comprises an input device 28, a link circuit 29 comprising a first bus 30 and a second bus 31, and an output device 32.

[0078] Each input 16, 17, 18 of the power converter 9 comprises a sensing module 303, 304, 305 comprising a voltage sensor and a current sensor measuring a voltage and a current of the input.

[0079] Each output 19, 20, 21 of the power converter 9 comprises a sensing module 306, 307, 308 comprising a voltage sensor and a current sensor measuring a voltage and a current of the output.

[0080] The first bus 30 may have a potential higher than the second bus 31.

[0081] The voltage between the first and second buses 30, 31 is noted VDC.

[0082] The link circuit 29 may further comprise a direct current DC link module 33 including two energy storage devices 33a, 33b connected in series.

[0083] A first end of a first energy storage device 33a is connected to the first bus 30, a first end of the second energy storage 33b is connected to the second bus 31, and the second end of the second energy storage 33b and the second end of the first energy storage 33a are connected to a connecting node 33c.

[0084] Each energy storage device 33a, 33b comprises for example, a capacitor.

[0085] The input device 28 comprises three inputs 34, 35, 36 and three outputs 37, 38, 39.

[0086] Each input 34, 35, 36 of the input device 28 is connected to a different input 16, 17, 18 of the power converter 9.

[0087] A first output 37 of the input device 28 is connected to the first bus 30, a second output 38 of the input device 28 is connected to the second bus 31 and the third output 39 is connected to the connecting node 33c of the DC link module 33.

[0088] The input device 28 further comprises identical input modules 40, 41, 42 intended to the supply the link circuit 29 with a continuous voltage from the phases of the supply grid.

[0089] Each input module 40, 41, 42 is connected to a different input 34, 35, 36 of the input device 28.

[0090] Each input module 40, 41, 42 comprises an input 40a, 41a, 42a connected to the input 34, 35, 36 of the input device 28, a first output 40b, 41b, 42b connected to the first output 37 of the input device 28, a second output 40c, 41c, 42c connected to the second output 38 of the input device 28, and a third output 40d, 41d, 42d connected to the third output 39 of the input device 28.

[0091] The output device 32 comprises three inputs 43, 44, 45 and three outputs 46, 47, 48.

[0092] A first input 43 of the output device 32 is connected to the first bus 30, a second input 44 of the output device 32 is connected to the second bus 31, and the third input 45 of the output device 32 is connected to the connecting node 33c of the DC link module 33.

[0093] A first output 46 of the output device 32 is connected to a first output 19 of the power converter 9, a second output 47 is connected to a second output 20 of the power converter 9, and a third output 48 is connected to the third output 21 of the power converter 9.

[0094] The output device 32 further comprises identical output modules 49, 50, 51 intended to supply the transformer 10 with alternative voltages from the link circuit 29.

[0095] Each output module 49, 50, 51 comprises a first input 49a, 50a, 51a connected to the first input 43 of the output device 32, a second input 49b, 50b, 51b connected to the second input 44 of the output device 32 and a third input 49c, 50c, 51c connected to the third input 45 of the output device 32.

[0096] Each output module 49, 50, 51 further comprises an output 49d, 50d, 51d connected to a different output 46, 47, 48 of the output device 32.

[0097] The input device 28 is intended to supply the link circuit 29 with a continuous voltage from the grid G.

[0098] The output device 32 is intended to supply with electrical energy the electrodes 2a, 2b, 2c through the transformer 10.

[0099] The energy storage devices 33a, 33b are designed to stabilize the continuous voltage and to

smooth the current variation on the buses **30, 31**.

[0100] The DC link module **33** may decouple the input device **28** and the output device **32** to avoid that electrical perturbations generated by the arc furnace **2** are rejected on the grid **G**.

[0101] In another embodiment, the link circuit **29** may not comprise the DC link module **33**, the third output **39** of the input device **28** and the third input **45** of the output device **32** being not connected.

[0102] As the input modules **40, 41, 42** are identical, FIG. **3** illustrates schematically a first example of the input module **40** comprising a first example of a hybrid modular multilevel converter HMMC.

[0103] The first embodiment of HMMC comprises a leg **52** comprising a first string **53** and a second string **54**.

[0104] The first string **53** comprises a first branch **55** and a second branch **56**.

[0105] Each branch **55, 56** comprises a first end **57, 58** and a second end **59, 60**.

[0106] The first end **57** of the first branch **55** is connected to the first output **40b** of the input module **40** and the second end **60** of the second branch **56** is connected to the second output **40c** of the input module **40**.

[0107] Each branch **55, 56** comprises a first set of input switches **61, 62, 63, 64** connected in series between the first and the second ends **57, 58, 59, 60** of the branch **55, 56**.

[0108] The first branch **55** further comprises a first connecting node **65** situated between two input switches **61, 62** of the first set of input switches.

[0109] The second branch **56** further comprises a second connecting node **66** situated between two input switches **63, 64** of the second set of input switches.

[0110] As represented, the number of input switches between the first connecting node **65** and the first end **57** of the branch **55, 56** may equal to the number of input switches between the first connecting node **65** and the second end **59**, and the number of input switches between the second connecting node **66** and the first end **58** of the branch **55, 56** may equal to the number of input switches between the second connecting node **66** and the second end **60**.

[0111] Each branch **55, 56** may comprise more than two input switches **61, 62, 63, 64**, the

[0112] two branches **55, 56** comprising a same number of input switches **61, 62, 63, 64**.

[0113] Each input switch **61, 62, 63, 64** comprises a first end **61a, 62a, 63a, 64a** and a second end **61b, 62b, 63b, 64b**.

[0114] The input switches **61, 62** of the first branch **55** are connected in series so that the

[0115] first end **61a** of the input switch **61** is connected to the first end **57** of the first branch **55**, the second end **61b** and the first end **62a** of the input switches **61, 62** are connected to the first connecting node **65**, and the second end **62b** of the input switch **62** is connected to the second end **59** of the first branch **55**.

[0116] The input switches **63, 64** of the second branch **56** are connected in series so that the first end **63a** of the input switch **63** is connected to the first end **58** of the second branch **56**, the second end **63b** and the first end **64a** of the input switches **63, 64** are connected to the second connecting node **66**, and the second end **64b** of the input switch **64** is connected to the second end **60** of the second branch **56**.

[0117] The first string **53** further comprises a third connecting node **67** connected to the second end **59** of the branch **55**, the first end **58** of the second branch **56**, and the third output **40d** of the input module **40**.

[0118] The first set of input switches **61, 62** connect the first bus **30** to the third connecting node **67**, and the second set of input switches **63, 64** connect the second bus **31** to the third connecting node **67**.

[0119] The second string **54** comprises a first portion **68**, a second portion **69**, and a fourth connecting node **70**.

[0120] The second string **54** may further comprise an inductor **71**.

[0121] Each first and second portions **68**, **69** comprises a first end **73**, **74**, a second end **75**, **76**.

[0122] The first portion **68** comprises a first set of input switching units **77** in series between the first and second ends **73**, **75** of the first portion **68**.

[0123] The second portion **69** comprises a second set of input switching units **77** in series between the first and second ends **74**, **76** of the second portion **69**.

[0124] Each set of input switching units **77** comprises a same number of input switching units **77**.

[0125] Each input switching unit **77** includes a plurality of controllable switches and energy storage means, the controllable switches being intended to bridge the energy storage means or to connect the energy storage means in each input switching unit **77**.

[0126] The input switching units **77** of each set are connected in series so that the first connection **78** of an input switching unit **77** is connected to a second connection **79** of an adjacent input switching unit **77**.

[0127] The first remaining connection **78** of the switching unit **77** of the first set is connected to the first end **73** and the second remaining connection **79** of the input switching unit **77** of the first set is connected to the second end **75**.

[0128] The first remaining connection of an input switching unit **77** is the first connection **78** of the input switching unit which is not connected to the first connection **78** or to the second connection **79** of another input switching unit **77**.

[0129] The second remaining connection **79** of an input switching unit **77** is the second connection **79** of the input switching unit which is not connected to the first connection **78** or to the second connection **79** of another input switching unit **77**.

[0130] The first end **73** of the first portion **68** is connected to the first connecting node **65** and the second end **75** is connected to a first end of the inductor **71**.

[0131] The first remaining connection **78** of the input switching unit **77** of the second set is connected to the first end **74** and the second remaining connection **79** of the input switching unit **77** of the second set is connected to the second end **76**.

[0132] The first set of input switching units **77** connects the first connecting node **66** to the fourth connecting node **70** and the second set of input switching units **77** connects the second connecting node **66** to the fourth connecting node **70**.

[0133] The first end **74** of the second portion **69** is connected to a second end of the inductor **71** and the second end **76** is connected to the second connecting node **66**.

[0134] The fourth connecting node **70** connects the input **40a** of the input module **40** to a mid-connecting node of the inductor **71**.

[0135] The inductor **71** permits to limit current peaks in the second string **54** and the phases of the grid G during commutations of the input switching units **77**.

[0136] In another embodiment, the second string **54** does not comprise the inductor **71**, the ends **74**, **75** of the first and second portions **68**, **69** and the input **40a** of the input module **40** being connected to the fourth connecting node **70**, the first portion **68** or the second portion **69** comprising an inductor to limit the current peaks in the second string **54** and the second string **54** comprises another inductor connecting the input **40a** of the input module **40** to the fourth connecting node **70** to limit the current peaks in the phase of the grid G.

[0137] In another embodiment, the second string **54** does not comprise the inductor **71**, the ends **74**, **75** of the first and second portions **68**, **69** and the input **40a** of the input module **40** are connected to the fourth connecting node **70**, each of the first portion **68** and the second portion **69** comprising an inductor to limit the current peaks in the second string **54** and the second string **54** may comprise another inductor connecting the input **40a** of the input module **40** to the fourth connecting node **70** to limit the current peaks in the phase of the grid G.

[0138] FIG. 4 illustrates schematically a second embodiment of the input module **40** comprising a second embodiment of the HMMC.

[0139] The second embodiment of HMMC comprises a leg **80** comprising a first string **81** and a

second string **82**.

[0140] The first string **81** comprises a first branch **83** and a second branch **84**.

[0141] Each branch **83**, **84** comprises a first end **85**, **86** and a second end **87**, **88**.

[0142] The first end **85** of the first branch **83** is connected to the first output **40b** of the input module **40** and the second end **88** of the second branch **84** is connected to the second output **40c** of the input module **40**.

[0143] The first branch **83** comprises a first input switch **89**, a first connecting node **90**, a first set **91** of input switching units **77** connected in series, and a first inductor **92**.

[0144] A first end **89a** of the first input switch **89** is connected to the first end **85** of the first branch **83** and a second end **89b** of the first input switch **89** is connected to the first connecting node **90**.

[0145] The input switching units **77** of the first set **91** are connected in series so that the first connection **78** of an input switching unit **77** is connected to a second connection **79** of an adjacent input switching unit **77**.

[0146] The first remaining connection **78** of the input switching unit **77** of the first set **91** is connected to the first connecting node **90** and the second remaining connection **79** of the input switching unit **77** of the first set **91** is connected to a first end of the first inductor **92**.

[0147] The second end of the first inductor **92** is connected to the second end **87** of the first branch **83**.

[0148] The second branch **84** comprises a second input switch **93**, a second connecting node **94**, a second set **95** of input switching units **77** connected in series, and a second inductor **96**.

[0149] A second end **93b** of the second input switch **93** is connected to the second end **88** of the second branch **84** and a first end **93a** of the second input switch **93** is connected to the second connecting node **94**.

[0150] The input switching units **77** of the second set **95** are connected in series so that the first connection **78** of an input switching unit **77** is connected to a second connection **79** of an adjacent input switching unit **77**.

[0151] The second remaining connection **79** of the input switching unit **77** of the second set **95** is connected to the second connecting node **94** and the first remaining connection **78** of the input switching unit **77** of the second set **95** is connected to a first end of the second inductor **96**.

[0152] The second end of the second inductor **96** is connected to the first end **86** of the second branch **84**.

[0153] The first and second sets **91**, **95** of input switching units **77** have a same number of input switching units **77**.

[0154] The first string **81** further comprises third connecting node **97** connected to the first end **86** of the second branch **84**, the second end of the first branch **83**, and the third input **40d** of the input module **40**.

[0155] The first input switch **89** connects the first bus **30** to the first connecting node **90** and the input switching units **77** connected in series of the first set **91** of input switching units **77**.

[0156] The input switching units **77** connected in series of the first set **91** of input switching units **77** connect the first connecting node **90** to the third connecting node **97**.

[0157] The second input switch **93** connects the second bus **31** to the second connecting node **94** and the second set **95** of input switching units **77**.

[0158] The input switching units **77** connected in series of the second set **95** of input switching units **77** connected in series connect the second connecting node **94** to the third connecting node **97**.

[0159] Each branch **83**, **84** may comprise more than one input switch, the input switches being connected in series between a connecting node and one end of the branch.

[0160] The first and second inductors **92**, **96** limit current peaks in the first string **81** and the phases of the grid **G** during commutations of the input switching units **77**.

[0161] The second string **82** comprises a first portion **98** and a second portion **99**.

[0162] Each portion **98, 99** comprises a first end **100, 101** and a second end **102, 103**.

[0163] The first portion **98** comprises a third input switch **104** comprising a first end **104a** connected to the first end **100** of said the first portion **98** and a second end **104b** connected to the second end **102** of said the first portion **98**.

[0164] The second portion **99** comprises a fourth input switch **105** comprising a first end **105a** connected to the first end **101** of the second portion **99** and a second end **105b** connected to the second end **103** of the second portion **99**.

[0165] In another embodiment, each first and second portion **98, 99** may comprise more than one input switch connected in series between both ends of said the first and second portion **98 99**, the number of input switches of the first and second portions being identical.

[0166] The second string **82** further comprises a fourth connecting node **106** connected to the second end **102** of the first portion **98**, the first end **101** of the second portion **99**, and the input **40a** of the input module **40**.

[0167] The first end **100** of the first portion **98** is connected to the first connecting node **90** and the second end **103** of the second portion **99** is connected to the second connecting node **94**.

[0168] The third input switch **104** connects the first connecting node **90** to the fourth connecting node **106** and the fourth input switch **105** connected in series connects the second connecting node **94** to the fourth connecting node **106**.

[0169] FIG. 5 illustrates schematically a third embodiment of the input module **40** comprising a third embodiment of the HMMC.

[0170] The third embodiment of HMMC comprises a leg **110** comprising a first string **111** and the second string **82** according to the second embodiment of HMMC illustrated in FIG. 4.

[0171] The first string **111** comprises a first branch **112** and a second branch **113**.

[0172] Each first and second branch **112, 113** comprises a first end **114, 115** and a second end **116, 117**.

[0173] The first end **114** of the first branch **112** is connected to the first output **40b** of the input module **40** and the second end **117** of the second branch **113** is connected to the second output **40c** of the input module **40**.

[0174] The first branch **112** comprises a first input switch **118**, a first connecting node **119**, a first inductor **120**, and a first set **121** of input switching units **77** connected in series.

[0175] A second end **118b** of the first input switch **118** is connected to the second end **116** of the first branch **112** and a first end **118a** of the first input switch **118** is connected to the first connecting node **119**.

[0176] The input switching units **77** of the first set **121** are connected in series so that the first connection **78** of an input switching unit **77** is connected to a second connection **79** of an adjacent input switching unit **77**.

[0177] The first remaining connection **78** of the input switching unit **77** of the first set **121** is connected to the first end **114** of the first branch **112** and the second remaining connection **79** of the input switching unit **77** of the first set **121** is connected to a first end of the first inductor **120**.

[0178] The second end of the first inductor **120** is connected to the first connecting node **119**.

[0179] The second branch **113** comprises a second input switch **122**, a second connecting node **123**, a second inductor **124**, and a second set **125** of input switching units **77** connected in series.

[0180] A first end **122a** of the second input switch **122** is connected to the first end **115** of the second branch **113** and a second end **122b** of the second input switch **122** is connected to the second connecting node **123**.

[0181] The input switching units **77** of the second set **125** are connected in series so that the first connection **78** of an input switching unit **77** is connected to a second connection **79** of an adjacent input switching unit **77**.

[0182] The first remaining connection **78** of the input switching unit **77** of the second set **125** is connected to a first end of the second inductor **124**.

[0183] A second end of the second inductor **124** is connected to the second connecting node **123**.

[0184] The second remaining connection **79** of the input switching unit **77** of the second set **125** is connected to the second end **117** of the second branch **113**.

[0185] The first and second sets **121**, **125** of input switching units **77** have a same number of input switching units **77**.

[0186] The first string **111** further comprises third connecting node **126** connected to the first end **115** of the second branch **113**, the second end **116** of the first branch **112**, and the third input **40d** of the input module **40**.

[0187] The first set **121** of input switching units **77** connected in series connects the first bus **30** to the first connecting node **119** and the first input switch **118** connects the first connecting node **119** to the third connecting node **126**.

[0188] The second set **125** of input switching units **77** in series connects the second bus **31** to the second connecting node **123** and the second input switch **122** connects the second connecting node **123** to the third connecting node **126**.

[0189] Each branch **112**, **113** may comprise more than one input switch, the input switches being connected in series between a connecting node and one end of the branch **112**, **113**.

[0190] The first and second inductors **120**, **124** limit current peaks in the first string **111** and the phases of the grid **G** during commutations of the input switching units **77**.

[0191] The first end **100** of the first portion **98** of the second string **82** is connected to the first connecting node **119** and the second end **103** of the second portion **99** of the second string **82** is connected to the second connecting node **123**.

[0192] FIG. **6** illustrates schematically a fourth embodiment of the input module **40** comprising an embodiment of a modular multi-level converter MMC.

[0193] The output module **40** comprises a string **130** including a first portion **131** and a second portion **132**.

[0194] The input module **40** comprises an inductor **133**.

[0195] Each portion **131**, **132** comprises a first end **134**, **135** and a second end **136**, **137**.

[0196] The first end **134** of the first portion **131** is connected to the first output **40b** of the input module **40** and the second end **137** of the second portion **132** is connected to the second output **40c** of the input module **40**.

[0197] The third output **40d** of the input module **40** is not connected.

[0198] The first portion **131** comprises a first set of input switching units **77** in series between the first and second ends **134**, **136** of said the first portion **131**.

[0199] The second portion **132** comprises a second set of input switching units **77** in series between the first and second ends **135**, **137** of the second portion **132**.

[0200] Each set of input switching units **77** comprises a same number of switching units **77**.

[0201] The input switching units **77** of each set are connected in series so that the first connection **78** of an input switching unit **77** is connected to a second connection **79** of an adjacent input switching unit **77**.

[0202] The first remaining connection **78** of the input switching unit **77** of the first set is connected to the first end **134** and the second remaining connection **79** of the input switching unit **77** of the first set is connected to the second end **136**.

[0203] The first remaining connection **78** of the input switching unit **77** of the second set is connected to the first end **135** and the second remaining connection **79** of the input switching unit **77** of the second set is connected to the second end **137**.

[0204] The string **130** further comprises a connecting node **138**.

[0205] The second end **136** of the first portion **131** is connected to a first end of the inductor **133** and the first end **135** of the second portion **132** is connected to the second end of the inductor **133**.

[0206] The connecting node **138** connects the input **40a** of the input module **40** to a mid-connecting node of the inductor **133**.

[0207] The first set of input switching units **77** is series connects the first bus **30** to the connecting node **138** and the second set of input switching units **77** in series connects the second bus **31** to the connecting node **138**.

[0208] The inductor **133** permits to limit current peaks in the string **130** and the phases of the grid **G** during commutations of the input switching units **77**.

[0209] In another embodiment, the string **130** does not comprise the inductor **133**, the ends **136**, **135** of the first and second portions **131**, **132** and the input **40a** of the input module **40** being connected to the connecting node **138**, the first portion **131** or the second portion **132** comprising an inductor to limit the current peaks in the string **130** and the string **130** comprises another inductor connecting the input **40a** of the input module **40** to the connecting node **130** to limit the current peaks in the phase of the grid **G**.

[0210] In another embodiment, the string **130** does not comprise the inductor **133**, the ends **136**, **135** of the first and second portions **131**, **132** and the input **40a** of the input module **40** are connected to the connecting node **138**, each of the first portion **131** and the second portion **132** comprising an inductor to limit the current peaks in the string **130** and the string **130** may comprise another inductor connecting the input **40a** of the input module **40** to the connecting node **138** to limit the current peaks in the phase of the grid **G**.

[0211] As the output modules **49**, **50**, **51** are identical, FIG. 7 illustrates schematically a first embodiment of the output module **49** comprising a first embodiment of the HMMC.

[0212] The first embodiment of the output module **49** comprises a leg **140** comprising a first string **141** and a second string **142**.

[0213] The first string **141** comprises a first branch **143** and a second branch **144**.

[0214] Each branch **143**, **144** comprises a first end **145**, **146** and a second end **147**, **148**.

[0215] The first end **145** of the first branch **143** is connected to the first input **49a** of the output module **49** and the second end **148** of the second branch **144** is connected to the second input **49b** of the output module **49**.

[0216] Each branch **143**, **144** comprises a first set of output switches **149**, **150**, **151**, **152** connected in series between the first and the second ends **145**, **146**, **147**, **148** of the branches **143**, **144**.

[0217] The first branch **143** further comprises a first connecting node **153** situated between two output switches **149**, **150** of the first set of output switches so that the number of output switches between the first connecting node **153** and the first end **145** of said the first branch **143** is equal to the number of output switches between the first connecting node **153** and the second end **147**.

[0218] The second branch **144** further comprises a second connecting node **154** situated between two output switches **151**, **152** of the second set of output switches so that the number of output switches between the second connecting node **154** and the first end **146** of the branch is equal to the number of output switches between the second connecting node **154** and the second end **148**.

[0219] Each branch **143**, **144** may comprise more than two output switches **149**, **150**, **151**, **152**, the two branches comprising a same number of output switches **149**, **150**, **151**, **152**.

[0220] Each output switch **149**, **150**, **151**, **152** comprises a first end **149a**, **150a**, **151a**, **152a** and a second end **149b**, **150b**, **151b**, **152b**.

[0221] The output switches **149**, **150** of the first branch **143** are connected in series so that the first end **150a** of the output switch **150** is connected to the first end **145** of the first branch **143**, the second end **149b** and the first end **150a** of the output switches **149**, **150** are connected to the first connecting node **153**, and the second end **150b** of the switch **150** is connected to the second end **147** of the first branch **143**.

[0222] The output switches **151**, **152** of the second branch **144** are connected in series so that the first end **151a** of the output switch **151** is connected to the first end **146** of the second branch **144**, the second end **151b** and the first end **152a** of the output switches **151**, **152** are connected to the second connecting node **154**, and the second end **152b** of the output switch **152** is connected to the second end **148** of the second branch **144**.

[0223] The first string **141** further comprises a third connecting node **155** connected to the second end **147** of the first branch **143**, the first end **146** of the second branch **144**, and the third input **49c** of the output module **49**.

[0224] The first set of output switches **149**, **150** in series connects the first bus **30** to the third connecting node **155**, and the second set of output switches **151**, **152** in series connects the second bus **31** to the third connecting node **155**.

[0225] The second string **142** comprises a first portion **156**, a second portion **157**, and a fourth connecting node **158**.

[0226] The second string **142** may further comprise an inductor **159a**.

[0227] Each first and second portion **156**, **157** comprises a first end **159**, **160**, a second end **161**, **162**.

[0228] The first portion **156** comprises a first set of output switching units **163** in series between the first and second ends **159**, **161** of the first portion **156**.

[0229] The second portion **157** comprises a second set of output switching units **163** in series between the first and second ends **160**, **162** of the second portion **157**.

[0230] Each set of output switching units **163** comprises a same number of output switching units **163**.

[0231] Each output switching unit **163** includes a plurality of controllable switches and energy storage means, the controllable switches being intended to bridge the energy storage means or to connect the energy storage means in each output switching unit **163**.

[0232] The output switching units **163** of each set are connected in series so that the first connection **164** of an output switching unit **163** is connected to a second connection **165** of an adjacent output switching unit **163**.

[0233] The first remaining connection of the output switching unit **163** of the first set is connected to the first end **159** and the second remaining connection **165** of the output switching unit **163** of the first set is connected to the second end **161**.

[0234] The first end **159** of the first portion **156** is connected to the first connecting node **153** and the second end **161** is connected to a first end of the inductor **159a**.

[0235] The first remaining connection of the output switching unit **163** of the second set is connected to the first end **160** and the second remaining connection of the output switching unit **163** of the second set is connected to the second end **162**.

[0236] The first set of output switching units **163** in series connects the first connecting node **153** to the fourth connecting node **158** and the second set of output switching units **163** in series connects the second connecting node **154** to the fourth connecting node **158**.

[0237] The first end **160** of the second portion **157** is connected to a second end of the inductor **159a** and the second end **162** is connected to the second connecting node **154**.

[0238] The fourth connecting node **158** connects the output **49d** of the output module **49** to a mid-connecting node of the inductor **159a**.

[0239] The inductor **159a** permits to limit current peaks in the second string **142** and the phases of the arc furnace **2** during commutations of the output switching units **163**.

[0240] In another embodiment, the second string **142** does not comprise the inductor **159a**, the ends **161**, **160** of the first and second portions **156**, **157** and the output **49d** of the output module **49** being connected to the fourth connecting node **158**, the first portion **156** or the second portion **157** comprising an inductor to limit the current peaks in the second string **142** and the second string **142** comprises another inductor connecting the output **49d** of the output module **49** to the fourth connecting node **159** to limit the current peaks in the phase of the arc furnace **2**.

[0241] In another embodiment, the second string **142** does not comprise the inductor **159a**, the ends **161**, **160** of the first and second portions **156**, **157** and the output **49d** of the output module **49** are connected to the fourth connecting node **158**, each of the first portion **156** and the second portion **157** comprising an inductor to limit the current peaks in the second string **142** and the second string

142 may comprise another inductor connecting the output **49d** of the output module **49** to the fourth connecting node **158** to limit the current peaks in the phase of the arc furnace **2**.

[0242] FIG. **8** illustrates schematically a second embodiment of the output module **49** comprising a second embodiment of the hybrid modular multilevel converter HMMC.

[0243] The second embodiment of the output module **49** comprises a leg **170** comprising a first string **171** and a second string **172**.

[0244] The first string **171** comprises a first branch **173** and a second branch **174**.

[0245] Each branch **173**, **174** comprises a first end **175**, **176** and a second end **177**, **178**.

[0246] The first end **175** of the first branch **173** is connected to the first input **49a** of the output module **49** and the second end **178** of the second branch **174** is connected to the second input **49b** of the output module **49**.

[0247] The first branch **173** comprises a first output switch **179**, a first connecting node **180**, a first set **181** of output switching units **163** connected in series, and a first inductor **182**.

[0248] A first end **179a** of the first output switch **179** is connected to the first end **175** of the first branch **173** and a second end **179b** of the first output switch **179** is connected to the first connecting node **180**.

[0249] The output switching units **163** of the first set **181** are connected in series so that the first connection **164** of an output switching unit **163** is connected to a second connection **165** of an adjacent output switching unit **163**.

[0250] The first remaining connection of the output switching unit **163** of the first set **181** is connected to the first connecting node **180** and the second remaining connection of the output switching unit **163** of the first set **181** is connected to a first end of the first inductor **182**.

[0251] The second end of the first inductor **182** is connected to the second end **177** of the first branch **173**.

[0252] The second branch **174** comprises a second output switch **183**, a second connecting node **184**, a second set **185** of output switching units **163** connected in series, and a second inductor **186**.

[0253] A second end **183b** of the second output switch **183** is connected to the second end **178** of the second branch **174** and a first end **183a** of the second output switch **183** is connected to the second connecting node **184**.

[0254] The output switching units **163** of the second set **185** are connected in series so that the first connection **164** of an output switching unit **163** is connected to a second connection **165** of an adjacent output switching unit **163**.

[0255] The second remaining connection **165** of the output switching unit **163** of the second set **185** is connected to the second connecting node **184** and the first remaining connection **164** of the output switching unit **163** of the second set **185** is connected to a first end of the second inductor **186**.

[0256] The second end of the second inductor **186** is connected to the first end **176** of the second branch **174**.

[0257] The first and second sets **181**, **185** of output switching units **163** have a same number of output switching units **163**.

[0258] The first string **171** further comprises a third connecting node **187** connected to the first end **176** of the second branch **174**, the second end **177** of the first branch **173** and the third input **49c** the output module **49**.

[0259] The first output switch **179** connects the first bus **30** to the first connecting node **180** and the first set **181** of output switching units **163** in series connects the first connecting node **180** to the third connecting node **187**.

[0260] The second output switch **183** connects the second bus **31** to the second connecting node **184** and the second set **185** of output switching units **163** in series connects the second connecting node **184** to the third connecting node **187**.

[0261] Each branch **173**, **174** may comprise more than one output switch, the output switches being

connected in series between a connecting node and one end of the branch.

[0262] The first and second inductors **182**, **186** limit current peaks in the first string **171** and the phases of the arc furnace **2** during commutations of the output switching units **163**. The second string **172** comprises a first portion **186** and a second portion **187**.

[0263] Each portion **186**, **187** comprises a first end **188**, **189** and a second end **190**, **191**.

[0264] The first portion **186** comprises a third output switch **192** comprising a first end **192a** connected to the first end **188** of the first portion **186** and a second end **192b** connected to the second end **190** of the first portion **186**.

[0265] The second portion **187** comprises a fourth output switch **193** comprising a first end **193a** connected to the first end **189** of the second portion **187** and a second end **193b** connected to the second end **191** of the second portion **187**.

[0266] In another embodiment, each first and second portion **186**, **187** may comprise more than one output switch connected in series between both ends of the portion **186**, **187**, the number of output switches of the first and second portions **186**, **187** being identical.

[0267] The second string **172** further comprises a fourth connecting node **194** connected to the second end **190** of the first portion **186**, the first end **189** of the second portion **187**, and the output **49d** of the output module **49**.

[0268] The first end **188** of the first portion **186** is connected to the first connecting node **180** and the second end **191** of the second portion **187** is connected to the second connecting node **184**.

[0269] The third output switch **192** connects the first connecting node **180** to the fourth connecting node **194** and the fourth output switch **193** in series connects the second connecting node **184** to the fourth connecting node **194**.

[0270] FIG. **9** illustrates schematically a third embodiment of the output module **49** comprising a third embodiment of the hybrid modular multilevel converter HMMC.

[0271] The third embodiment of the output module **49** comprises a leg **195** comprising a first string **196** and the second string **172** according to the second embodiment of the hybrid modular multilevel converter illustrated in FIG. **8**.

[0272] The first string **196** comprises a first branch **197** and a second branch **198**.

[0273] Each branch **197**, **198** comprises a first end **199**, **200** and a second end **201**, **202**.

[0274] The first end **199** of the first branch **197** is connected to the first input **49a** of the output module **49** and the second end **202** of the second branch **198** is connected to the second input **49b** of the output module **49**.

[0275] The first branch **197** comprises a first output switch **203**, a first connecting node **204**, a first inductor **205**, and a first set **206** of output switching units **163** connected in series.

[0276] A second end **203b** of the first output switch **203** is connected to the second end **201** of the first branch **197** and a first end **203** of the first output switch **203** is connected to the first connecting node **204**.

[0277] The output switching units **163** of the first set **206** are connected in series so that the first connection **164** of an output switching unit **163** is connected to a second connection **165** of an adjacent output switching unit **163**.

[0278] The first remaining connection **164** of the output switching unit **163** of the first set **206** is connected to the first end **199** of the first branch **197** and the second remaining connection **165** of the output switching unit **163** of the first set **206** is connected to a first end of the first inductor **205**.

[0279] The second end of the first inductor **205** is connected to the first connecting node **204**.

[0280] The second branch **198** comprises a second output switch **207**, a second connecting node **208**, a second inductor **209**, and a second set **210** of input switching units **163** connected in series.

[0281] A first end **207a** of the second output switch **207** is connected to the first end **200** of the second branch **198** and a second end **207b** of the second output switch **207** is connected to the second connecting node **208**.

[0282] The output switching units **163** of the second set **210** are connected in series so that the first

connection **164** of an output switching unit **163** is connected to a second connection **165** of an adjacent switching unit **163**.

[0283] The first remaining connection **164** of the output switching unit of the second set **210** is connected to a first end of the second inductor **209**.

[0284] A second end of the second inductor **209** is connected to the second connecting node **208**.

[0285] The second remaining connection **165** of the output switching unit **163** of the second set **210** is connected to the second end **202** of the second branch **198**.

[0286] The first and second sets **206**, **210** of output switching units **163** have a same number of output switching units **163**.

[0287] The first string **196** further comprises a third connecting node **211** connected to the first end **200** of the second branch **198**, the second end **201** of the first branch **197** and the third input **49c** of the output module **49**.

[0288] The first set **206** of output switching units **163** in series connects the first bus **30** to the first connecting node **204** and the first output switch **203** connects the first connecting node **204** to the third connecting node **211**.

[0289] The second set **210** of output switching units **163** in series connects the second bus **31** to the second connecting node **208** and the second input switch **207** connects the second connecting node **208** to the third connecting node **211**.

[0290] Each branch **197**, **198** may comprise more than one output switch, the output switches being connected in series between a connecting node and one end of the branch.

[0291] The first and second inductors **205**, **209** limit current peaks in the first string **196** and the phases of the arc furnace **2** during commutations of the output switching units **163**.

[0292] The first end **188** of the first portion **186** of the second string **172** is connected to the first connecting node **204** and the second end **191** of the second portion **187** of the second string **172** is connected to the second connecting node **208**.

[0293] FIG. **10** illustrates schematically a fourth embodiment of the output module **49** comprising an embodiment of a modular multi-level converter MMC.

[0294] The output module **49** comprises a string **220** including a first portion **221** and a second portion **222**.

[0295] The output module **49** comprises an inductor **223**.

[0296] Each portion **221**, **222** comprises a first end **224**, **225**, a second end **226**, **227**.

[0297] The first end **224** of the first portion **221** is connected to the first input **49a** of the output module **49** and the second end **227** of the second portion **222** is connected to the second input **49b** of the output module **49**.

[0298] The input **49c** of the output module **49** is not connected.

[0299] The first portion **221** comprises a first set of output switching units **163** in series between the first and second ends **224**, **226** of the first portion **221**.

[0300] The second portion **222** comprises a second set of output switching units **163** in series between the first and second ends **225**, **227** of the second portion **222**.

[0301] Each set of output switching units **163** comprises a same number of output switching units **163**.

[0302] The output switching units **163** of each set are connected in series so that the first connection **164** of an output switching unit **163** is connected to a second connection **165** of an adjacent output switching unit **163**.

[0303] The first remaining connection **164** of the output switching unit **163** of the first set is connected to the first end **224** and the second remaining connection **165** of the output switching unit **163** of the first set is connected to the second end **226**.

[0304] The first remaining connection **164** of the output switching unit **163** of the second set **222** is connected to the first end **225** and the second remaining connection **165** of the output switching unit **163** of the second set is connected to the second end **227**.

[0305] The string **220** further comprises a connecting node **228**.

[0306] The second end **226** of the first portion **221** is connected to a first end of the inductor **223** and the first end **225** of the second portion **222** is connected to the second end of the inductor **223**.

[0307] The connecting node **228** connects the output **49d** of the output module **49** to a mid-connecting node of the inductor **223**.

[0308] The first set of output switching units **163** in series connects the first bus **30** to the connecting node **228** and the second set of output switching units **163** in series connects the second bus **31** to the connecting node **228**.

[0309] The inductor **223** permits to limit current peaks in the string **220** and the phases of the grid G during commutations of the output switching units **163**.

[0310] In another embodiment, the string **220** does not comprise the inductor **223**, the ends **226**, **225** of the first and second portions **221**, **222** and the output **49d** of the output module **49** being connected to the connecting node **228**, the first portion **221** or the second portion **222** comprising an inductor to limit the current peaks in the string **220** and the string **220** comprises another inductor connecting the output **49d** of the output module **49** to the connecting node **228** to limit the current peaks in the phase of the arc furnace **2**.

[0311] In another embodiment, the string **220** does not comprise the inductor **223**, the ends **226**, **225** of the first and second portions **221**, **222** and the output **49d** of the output module **49** are connected to the connecting node **228**, each of the first portion **221** and the second portion **222** comprising an inductor to limit the current peaks in the string **220** and the string **220** may comprise another inductor connecting the output **49d** of the output module **49** to the connecting node **228** to limit the current peaks in the phase of the arc furnace **2**.

[0312] FIG. **11** illustrates schematically a first embodiment of the input switches **61**, **62**, **63**, **64**, **89**, **93**, **104**, **105**, **118**, **122** and the output switches **149**, **150**, **151**, **152**, **179**, **183**, **192**, **193**, **203**, **207**.

[0313] Each switch **61**, **62**, **63**, **64**, **89**, **93**, **104**, **105**, **118**, **122**, **149**, **150**, **151**, **152**, **179**, **183**, **192**, **193**, **203**, **207** of the input and output switches comprises a diode **230** extending between the first and the second ends of the switch **61**, **62**, **63**, **64**, **89**, **93**, **104**, **105**, **118**, **122**, **149**, **150**, **151**, **152**, **179**, **183**, **192**, **193**, **203**, **207** so that the anode of the diode **230** is connected to the second end **61b**, **62b**, **63b**, **64b**, **89b**, **93b**, **104b**, **105b**, **118b**, **122b**, **149b**, **150b**, **151b**, **152b**, **179b**, **183b**, **192b**, **193b**, **203b**, **207b** of the switch and the cathode is connected to the first end **61a**, **62a**, **63a**, **64a**, **89a**, **93a**, **104a**, **105a**, **118a**, **122a**, **149a**, **150a**, **151a**, **152a**, **179a**, **183a**, **192a**, **193a**, **203a**, **207a** of the switch.

[0314] As the diode **230** does need to be controlled by the control circuit **11**, the first embodiment of the switch **61**, **62**, **63**, **64**, **89**, **93**, **104**, **105**, **118**, **122**, **149**, **150**, **151**, **152**, **179**, **183**, **192**, **193**, **203**, **207** comprising the diode **230** simplify the implementation of the input switches in the input modules **40**, **41**, **42** and in the implementation of the output switches in the output modules **49**, **50**, **51**.

[0315] Each switch **61**, **62**, **63**, **64**, **89**, **93**, **104**, **105**, **118**, **122**, **149**, **150**, **151**, **152**, **179**, **183**, **192**, **193**, **203**, **207** may comprise a plurality of diodes **230** in series so that the anode of one diode is connected to the cathode of another diode.

[0316] When the input switches **61**, **62**, **63**, **64**, **89**, **93**, **104**, **105**, **118** and **122** according to the first embodiment comprising the diode **230** are implemented in input modules comprising an HMMC as illustrated in FIGS. **3** to **5** according to the first, second and third embodiments of the input modules or when the output switches **149**, **150**, **151**, **152**, **179**, **183**, **192**, **193**, **203** and **207** according to the first embodiment comprising the diode **230** are implemented in output modules comprising an HMMC as illustrated in FIGS. **6** to **8** according to the first, second and third embodiments of the output modules, the diode **230** is designed to support a voltage at least equal to $V_{DC}/2$.

[0317] Each input or output switch **61**, **62**, **63**, **64**, **89**, **93**, **104**, **105**, **118**, **122**, **149**, **150**, **151**, **152**, **179**, **183**, **192**, **193**, **203**, **207** may comprise a plurality of diodes **230** in series so that the anode of

one diode is connected to the cathode of another diode.

[0318] FIG. 12 illustrates schematically a first embodiment of the input switches **61, 62, 63, 64, 89, 93, 104, 105, 118, 122** and the output switches **149, 150, 151, 152, 179, 183, 192, 193, 203, 207**.

[0319] Each switch **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** of the input and output switches comprises the diode **230** extending between the first and the second ends of the switch **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** so that the anode of the diode **230** is connected to the second end **61b, 62b, 63b, 64b, 89b, 93b, 104b, 105b, 118b, 122b, 149b, 150b, 151b, 152b, 179b, 183b, 192b, 193b, 203b, 207b** of the switch and the cathode is connected to the first end **61a, 62a, 63a, 64a, 89a, 93a, 104a, 105a, 118a, 122a, 149a, 150a, 151a, 152a, 179a, 183a, 192a, 193a, 203a, 207a** of the switch.

[0320] Each switch further comprises a gate turn off thyristor **231**.

[0321] The gate of the gate turn off thyristor **231** is connected to the control circuit **11**, the anode of the gate turn off thyristor **231** is connected to the cathode of the diode **230**, and the cathode of the gate turn off thyristor **231** is connected to the anode of the diode **230**.

[0322] Each switch may comprise a plurality of gate of the gate turn off thyristors **231** in series so that the anode of one thyristor is connected to the cathode of another thyristor.

[0323] When the input switches according to the second embodiment of the switches comprising the diode **230** and the gate turn off thyristor **230** are implemented in input modules comprising an HMMC as illustrated in FIGS. 3 to 5 according to the first, second and third embodiments of the input modules or when the output switches according to the second embodiment of the switches are implemented in output modules comprising an HMMC as illustrated in FIGS. 6 to 8 according to the first, second and third embodiments of the output modules, the diode **230** and the gate turn off thyristors **231** are designed to support a voltage at least equal to $VDC/2$.

[0324] Each input or output switch **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** may comprise a plurality of gate turn off thyristors **231** in parallel with the diode **230** in series so that the anode of one diode is connected to the cathode of another diode.

[0325] FIG. 13 illustrates schematically a third embodiment of the input switches **61, 62, 63, 64, 89, 93, 104, 105, 118, 122** and the output switches **149, 150, 151, 152, 179, 183, 192, 193, 203, 207**.

[0326] Each switch **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** of the input and output switches comprises the diode **230** extending between the first and the second ends of the switch **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** so that the anode of the diode **230** is connected to the second end **61b, 62b, 63b, 64b, 89b, 93b, 104b, 105b, 118b, 122b, 149b, 150b, 151b, 152b, 179b, 183b, 192b, 193b, 203b, 207b** of the switch and the cathode is connected to the first end **61a, 62a, 63a, 64a, 89a, 93a, 104a, 105a, 118a, 122a, 149a, 150a, 151a, 152a, 179a, 183a, 192a, 193a, 203a, 207a** of the switch.

[0327] Each switch **61, 62, 63, 64, 89, 93, 104, 105, 118, 122, 149, 150, 151, 152, 179, 183, 192, 193, 203, 207** further comprises a field effect transistor **232** according to an embodiment.

[0328] The field effect transistor **232** may be an insulated gate bipolar transistor for example using silicon-base semiconductors such as insulated gate bipolar transistor IGBT or injection-enhanced gate transistor IEGT, or using silicon carbide semiconductors such as metal-oxide-semiconductor field-effect transistor MOSFET.

[0329] The gate of the field effect transistor **232** is connected to the control circuit **11**, the drain of the transistor **232** is connected to the cathode of the diode **230** and the source of the field effect transistor **232** is connected to the anode of the diode **230**.

[0330] Each switch may comprise a plurality of field effect transistors **232** and diodes **230** in parallel in series so that the drain of one field effect transistor **232** is connected to the source of

another field effect transistor **232**.

[0331] When the input switches **61, 62, 63, 64, 89, 93, 104, 105, 118** and **122** according to the third embodiment are implemented in input modules comprising an HMMC as illustrated in FIGS. **3** to **5** according to the first, second and third embodiments of the input modules or when the output switches **149, 150, 151, 152, 179, 183, 192, 193, 203** and **207** according to the third embodiment are implemented in output modules comprising an HMMC as illustrated in FIGS. **6** to **8** according to the first, second and third embodiments of the output modules, the diode **230** is designed to support a voltage at least equal to $V_{DC}/2$.

[0332] The inputs modules **40, 41, 42** may comprise one embodiment of the input switches illustrated in FIGS. **11, 12, 13**, and the output modules **49, 50, 51** may comprise the same or another embodiment of the output switches illustrated in FIGS. **11, 12, 13**.

[0333] FIG. **14** illustrates schematically a first embodiment of the input switching unit **77** and the output switching unit **163**.

[0334] Each switching unit of the input switching unit **77** and the output switching unit **163** comprises two controllable switches **233, 234** and an energy storage means **235** comprising a first end **235a** and a second end **235b**.

[0335] The energy storage means **235** comprises for example, a capacitor.

[0336] Each controllable switch **233, 234** comprises a first end **233a, 234a**, a second end **233b, 234b**, and a control input **233c, 234c** connected to the control circuit **11**.

[0337] The first end **233a** of a first controllable switch **233** is connected to the first end **235a** of the energy storage means **235**.

[0338] The second end **233b** of the first controllable switch **233** and the first end **234a** of the second controllable switch **234** are connected to the first end **78, 164** of the input switching unit **77** or output switching unit **163**.

[0339] The second end **234b** of the second controllable switch **234** is connected to the second end **235b** of the energy storage means **235** and to the second end **79, 165** of the input switching unit **77** or output switching unit **163**.

[0340] As the controllable switches **233, 234** are identical, only the first switch **233** is detailed.

[0341] The first switch **233** comprises a transistor **236** and a diode **237**.

[0342] According to an embodiment, the transistor **236** may be a field effect transistor.

[0343] The transistor **236** may be an insulated gate bipolar transistor for example, using silicon-base semiconductors such as insulated gate bipolar transistor IGBT or injection-enhanced gate transistor IEGT, or using silicon carbide semiconductors such as metal-oxide-semiconductor field-effect transistor MOSFET.

[0344] The gate of the transistor **236** is connected to the control input **233c**, the drain of the transistor **236** and the cathode of the diode **237** are connected to the first end **233a**, and the source of the transistor **236** and the anode of the diode **237** are connected to the second end **233b**.

[0345] FIG. **15** illustrates schematically a second embodiment of the input switching unit **77** and the output switching unit **163**.

[0346] Each switching unit of the input switching unit **77** and the output switching unit **163** comprises four controllable switches **238, 239, 240, 241** identical to the controllable switches **233, 234** of the first embodiment illustrated in FIG. **14**.

[0347] Each controllable switches **238, 239, 240, 241** comprises a first end **238a, 239a, 240a, 241a**, a second end **238b, 239b, 240b, 241b**, and a control input **238c, 239c, 240c, 241c** connected to the control circuit **11**.

[0348] Each switching unit **77, 163** further comprises the energy storage means **235**

[0349] The first end **238a** of a first controllable switch **238** and the first end **239a** of a second controllable switch **239** are connected to the first end **235a** of the energy storage means **235**.

[0350] The second end **240b** of a third controllable switch **240** and the second end **241b** of the fourth controllable switch **241** are connected to the second end **235b** of the energy storage means

235.

[0351] The second end **238b** of the first controllable switch **238** and the first end **240a** of the third controllable switch **240** are connected to the first end **78, 164** of the input switching unit **77** or output switching unit **163**.

[0352] The second end **239b** of the second controllable switch **239** and the first end **241a** of the fourth controllable switch **241** are connected to the second end **79, 165** of the input switching unit **77** or output switching unit **163**.

[0353] The input switching units **77** and the output switching units **163** according to the first embodiment illustrated in FIG. **14** have two states, each state delivering a different voltage value on the ends of the input switching units **77** and the output switching units **163**.

[0354] For example, in a first state, the input and output switching units **77, 163** deliver a positive compensating voltage V_{cc} , and in a second state a nil voltage.

[0355] The input switching units **77** and the output switching units **163** according to the second embodiment illustrated in FIG. **15** have three states, each state delivering a different voltage value on the ends of the input switching units **77** and the output switching units **163**.

[0356] For example, in a first state, the input and output switching units **77, 163** deliver a positive voltage V_{cc} , in a second state a nil voltage, and in a third state a negative compensating voltage- V_{cc} .

[0357] The embodiment of the input switching units **77** of inputs modules **40, 41, 42** and the embodiment of the output switching units **163** of the output modules **49, 50, 51** may be identical or different.

[0358] FIG. **16** illustrates a second embodiment of the arc furnace facility.

[0359] The arc furnace facility differs from the arc furnace facility illustrated in FIG. **1** in that the arc furnace facility comprises a direct voltage arc furnace **250**.

[0360] The direct voltage arc furnace **250** comprises the set of electrodes comprising only one electrode **250a**.

[0361] To supply the direct voltage arc furnace **250**, the power supply system **1** further comprises a rectifier **251**.

[0362] The rectifier **251** comprises a plurality of inputs **252, 253, 254**, being connected the phase **25, 26, 27** of the secondary circuit of the transformer **10**.

[0363] The rectifier **251** further comprises a first output **255** connected to an enclosure **250b** of the arc furnace **250** and a second output **256** connected to the electrode **250a** of the arc furnace **250**.

[0364] The rectifier **251** comprises the same number of identical legs **257, 258, 259** as the number of phases of the secondary circuit of the transformer **10**, each leg **257, 258, 259** being connected to a different phase.

[0365] Each leg **257, 258, 259** comprises a first connection **257a, 258a, 259a**, a second connection **257b, 258b, 259b**, and a third connection **257c, 258c, 259c**.

[0366] The first connection **257a, 258a, 259a** of each leg **257, 258, 259** is connected to the first output **255** of the rectifier **251**, the second connection **257b, 258b, 259b** of each leg **257, 258, 259** is connected to the second output **256** of the rectifier **251**, and the third connection **257c, 258c, 259c** is connected to a different input **252, 253, 254** of the rectifier **251**.

[0367] As the legs **257, 258, 259** are identical, only a first leg **257** is detailed.

[0368] The first leg **257** comprises two identical power diodes **260, 261** connected in series.

[0369] The cathode of a first power diode **260** is connected to the first connection **257a**, the anode of the first power diode **260** and the cathode of the second power diode **261** are connected to the third connection **257c**, and the anode of the second power diode **261** is connected to the second connection **257b**.

[0370] The rectifier **251** rectifies the voltage delivered by each phase of the secondary circuit to supply the electrode **250a** of the direct voltage arc furnace **250**.

[0371] In another embodiment, the rectifier **251** may comprise an active rectifier made from

controlled semiconductor, for example transistors and/or gate turn off thyristors.

[0372] The implementation of a HMMC according to the embodiments described above in input and output modules permits to reduce the volume, the size and the weight of the modules, and to increase power density and efficiency of the modules.

[0373] A first embodiment of a method implementing the power supply system **1** is presented.

[0374] The control circuit **11** opens the first disconnecting means **13**, the third disconnecting means **270** and the first isolation switch **15A**, and closes the second disconnecting means **14**, the second isolation switch **15B** and if applicable the third isolation switch **15C**.

[0375] As the first disconnecting means **13**, the third disconnecting means **270** and the first isolation switch **15A** are open, no current flows through the first disconnecting means **13**, the third disconnecting means **270** and the first isolation switch **15A**.

[0376] As the second disconnecting means **14** are closed, currents flow from the outputs **19, 20, 21** of the power converter **9** to the primary circuit of the transformer **10**.

[0377] The control circuit **11** controls the input modules **40, 41, 42** and the outputs modules **49, 50, 51** to supply the set of electrodes and to stabilize the current and the voltage delivered by the grid G to reduce reactions in the grid G generated by the arc furnace **2, 250**.

[0378] The control circuit **11** controls independently each input switching unit **77** and each output switching unit **163**, and if applicable each input switch and each output switch to supply the set of electrodes and to stabilize the current and the voltage delivered by the grid G to reduce reactions in the grid G.

[0379] In a second embodiment of the method implementing the power supply system **1**, the control circuit **11** closes the first disconnecting means **13** so that currents flow from the grid G to the primary circuit through the first disconnecting means **13**, the second isolation switch **15B** so that currents flow through the second isolation switch **15B** and if applicable the third isolation switch **15C** so that currents flow through the third isolation switch **15C**, and opens the second disconnecting means **14** and the third disconnecting means **270** so that no current flows from the outputs of the power converter **9** to the primary circuit, the primary circuit being supplied directly by the grid G.

[0380] The output switching unit **163** is idle and if applicable the output switches are open.

[0381] The control circuit **11** controls independently each input switching unit **77** of the input modules to stabilize the current and the voltage delivered by the grid G to reduce reactions in the grid G.

[0382] The input modules of the power converter **9** act as a static compensator STATCOM to reduce the reactions on the grid G.

[0383] When the power converter **9** comprises the input modules **40, 41, 42** according to the first embodiment illustrated in FIG. 3, in this second embodiment of the method, the control circuit **11** further closes the input switch **61** of the first set of input switches between the first end **57** of the first branch **55** and the first connecting node **65** of the first branch **55**, and the input switch **64** of the second set of input switches between the second end **60** of the second branch **56** and the second connecting node **66** of the second branch **56**, and the control circuit **11** further opens the input switch **62** of the first set of input switches between the first connecting node **65** of the first branch **55** and the third connecting node **67** of the first string **53** and, the input switch **63** of the second set of input switches between the second connecting node **66** of the second branch **56** and the third connecting node **67** of the first string **53**.

[0384] In an embodiment, the control circuit **11** further opens the input switch **61** of the first set of input switches between the first end **57** of the first branch **55** and the first connecting node **65** of the first branch **55**, and the input switch **64** of the second set of input switches between the second end **60** of the second branch **56** and the second connecting node **66** of the second branch **56**, and the control circuit **11** further closes the input switch **62** of the first set of input switches between the first connecting node **65** of the first branch **55** and the third connecting node **67** of the first string **53**

and, the input switch **63** of the second set of input switches between the second connecting node **66** of the second branch **56** and the third connecting node **67** of the first string **53**.

[0385] When the power converter **9** comprises the input modules **40, 41, 42** according to the second embodiment illustrated in FIG. **4**, in this second embodiment of the method, the control circuit **11** further opens the first input switch **89** and the second input switch **93**, and further closes the third input switch **104** and the fourth input switch **105**.

[0386] When the power converter **9** comprises the input modules **40, 41, 42** according to the third embodiment illustrated in FIG. **5**, in this second embodiment of the method, the control circuit **11** further opens the first input switch **118** and the second input switch **122**, and further closes the third input switch **104** and the fourth input switch **105**.

[0387] A third embodiment of the method implementing the power supply system **1** differs from the second embodiment of the method in that the control circuit **11** further closes the third disconnecting means **270**.

[0388] The power converter **9** acting as a STATCOM injects power in the grid G through the third disconnecting means **270**.

[0389] In a fourth embodiment of the method implementing the power supply system **1**, the control circuit **11** opens the first disconnecting means **13**, the third disconnecting means **270** so that currents flow from the grid G to the inputs **16, 17, 18** of the power converter **9** and so that no current flows directly from the grid G to the second disconnecting means **14**, the second isolation switch **15B** and if applicable the third isolation switch **15C**, and closes the second disconnecting means **14** and the first isolation switch **15A** so that electrode(s) of the arc furnace **2, 250** is directly supplied by the power converter **9**, the transformer **10** being bypassed to simplify the control of the power converter **9** to supply the arc furnace **2, 250**, and to improve the reliability and the efficiency of the power supply system **1**.

[0390] The first and second isolation switches **15A, 15B** permit to bypass and isolate the transformer **10**.

[0391] The third isolation switch **15C** reinforces the isolation of the bypassed transformer **10** electrically and mechanically.

Claims

1. A power supply system for arc furnace, the power supply system including: a power converter comprising an input to be connected to polyphase supply grid and an output, and a polyphase transformer comprising a primary circuit and a secondary circuit, a first disconnecting means, and a second disconnecting means, the power supply system further comprises a first isolation switch and a second isolation switch, the first disconnecting means connecting the input of the power converter to a first end of the first isolation switch, the second disconnecting means connecting the output of the power converter to the first end of the first isolation switch, the second end of the first isolation switch being intended to be connected to at least one electrode of the arc furnace, the second isolation switch connecting the first end of the first isolation switch to the primary circuit and the secondary circuit being connected to the second end of the first isolation switch, a third disconnecting means connecting the input of the power converter to the output of the power converter, and a control circuit configured to: control the power converter to supply the electrode and stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, the first isolation switch is open and the second isolation switch is closed or the first isolation switch is closed and the second isolation switch is open, and the third disconnecting means are open, and control the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second isolation switch is closed, and the

third disconnecting means are open or closed.

2. The power supply system according to claim 1, wherein the power converter comprises an input device, a link circuit comprising a first bus and a second bus, and an output device, wherein: the input device comprises a plurality of input modules, each input module being connected to the first bus and the second bus and being intended to be connected to a different phase of the supply grid to supply the link circuit with a continuous voltage from the phase of the supply grid, the output device comprises a plurality of output modules, each output module being connected to the first bus and second bus and to the primary circuit to supply the transformer with an alternative voltage from the link circuit.

3. The power supply system according to claim 2, wherein each input module comprises: a leg including: a first string comprising: a first set of input switches connected in series, a second set of input switches connected in series, a first connecting node, a second connecting node, and a third connecting node, the first set of input switches connecting the first bus to the third connecting node, and the first connecting node being disposed between two input switches of the first set of input switches, the second set of input switches connecting the second bus to the third connecting node, and the second connecting node being disposed between two input switches of the second set of input switches, a second string comprising: a first set of input switching units in series, a second set of input switching units in series, and a fourth connecting node, the first set of input switching units connecting the first connecting node to the fourth connecting node, the second set of input switching units connecting the second connecting node to the fourth connecting node, and the fourth connecting node being intended to be connected to the phase of the supply grid, the control circuit being configured to control the input switches, the input switching units, and the output modules, the third connecting nodes of the input modules being connected together.

4. The power supply system according to claim 2, wherein each input module comprises: a leg including: a first string comprising: a first input switch, a second input switch, a first connecting node, a second connecting node, a third connecting node, a first set of input switching units connected in series, and a second set of input switching units connected in series, the first input switch connecting the first bus to the first connecting node, and the first set of input switching units connecting the first connecting node to the third connecting node, the second input switch connecting the second bus to the second connecting node, and the second set of input switching units connecting the second connecting node to the third connecting node, a second string comprising: a third input switch, a fourth input switch, and a fourth connecting node, the third input switch connecting the first connecting node to the fourth connecting node, and the fourth input switch connecting the second connecting node to the fourth connecting node, the fourth connecting node being intended to be connected to the phase of the supply grid, the control circuit being configured to control the input switches, the input switching units, and the output modules, the third connecting nodes of the input modules being connected together.

5. The power supply system according to claim 2, wherein each input module comprises: a leg including: a first string comprising: a first input switch, a second input switch, a first connecting node, a second connecting node, a third connecting node, a first set of input switching units connected in series, and a second set of input switching units connected in series, the first set of input switching units connecting the first bus to the first connecting node, and the first input switch connecting the first connecting node to the third connecting node, the second set of input switching units connecting the second bus to the second connecting node, and the second input switch connecting the second connecting node to the third connecting node, a second string comprising: a third input switch, a fourth input switch, and a fourth connecting node, the third input switch connecting the first connecting node to the fourth connecting node, and the fourth input switch connecting the second connecting node to the fourth connecting node, the fourth connecting node being intended to be connected to the phase of the supply grid, the control circuit being configured to control the input switches, the input switching units, and the output modules, the third

connecting nodes of the input modules being connected together.

6. The power supply system according to claim 2, wherein each input module comprises a string comprising: a connecting node being intended to be connected to the phase of the supply grid, a first set of input switching units connected in series, and a second set of input switching units connected in series, the first set of input switching units connecting the first bus to the connecting node, and the second set of input switching units connecting the second bus to the connecting node, and the control circuit being configured to control the input switching units, and the output modules.

7. The power supply system according to claim 6, wherein each output module comprises: a leg including: a first string comprising: a first set of output switches connected in series, a second set of output switches connected in series, a first connecting node, a second connecting node, and a third connecting node, the first set of output switches connecting the first bus to the third connecting node, and the first connecting node being disposed between two output switches of the first set of output switches, the second set of output switches connecting the second bus to the third connecting node, and the second connecting node being disposed between two output switches of the second set of output switches, a second string comprising: a first set of output switching units in series, a second set of output switching units in series, and a fourth connecting node, the first set of output switching units connecting the first connecting node to the third connecting node, the second set of output switching units connecting the second connecting node to the fourth connecting node, and the fourth connecting node being intended to be connected to a phase of the arc furnace, the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

8. The power supply system according to claim 6, wherein each output module comprises: a leg including: a first string comprising: a first output switch, a second output switch, a first connecting node, a second connecting node, a third connecting node, a first set of output switching units connected in series, and a second set of output switching units connected in series, the first output switch connecting the first bus to the first connecting node, and the first set of output switching units connecting the first connecting node to the third connecting node, the second output switch connecting the second bus to the second connecting node, and the second set of output switching units connecting the second connecting node to the third connecting node, a second string comprising: a third output switch, a fourth output switch, and a fourth connecting node, the third output switch connecting the first connecting node to the fourth connecting node, and the fourth output switch connecting the second connecting node to the fourth connecting node, the fourth connecting node being intended to be connected to a phase of the arc furnace, the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

9. The power supply system according to claim 6, wherein each output module comprises: a leg including: a first string comprising: a first output switch, a second output switch, a first connecting node, a second connecting node, a third connecting node, a first set of output switching units connected in series, and a second set of output switching units connected in series, the first set of output switching units connecting the first bus to the first connecting node, and the first output switch connecting the first connecting node to the third connecting node, the second set of output switching units connecting the second bus to the second connecting node, and the second output switch connecting the second connecting node to the third connecting node, a second string comprising: a third output switch, a fourth output switch, and a fourth connecting node, the third output switch connecting the first connecting node to the fourth connecting node, and the fourth output switch connecting the second connecting node to the fourth connecting node, the fourth connecting node being intended to be connected to a phase of the arc furnace, the control circuit being configured to control the output switches, the output switching units, and the output modules, the third connecting nodes of the output modules being connected together.

- 10.** The power supply system according to claim 6, wherein each output module comprises a string comprising: a connecting node being intended to be connected to a phase of the arc furnace, a first set of output switching units connected in series, and a second set of output switching units connected in series, the first set of output switching units connecting the first bus to the connecting node, and the second set of output switching units connecting the second bus to the connecting node, the control circuit being configured to control the output switching units and the output modules.
- 11.** The power supply system according to claim 10, further comprising a third isolation switch, the third isolation switch connecting the secondary circuit to the second end of the first isolation switch, the control circuit being configured to: control the power converter to supply the electrode and stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, the first isolation switch is open and the second and third isolation switches are closed or the first isolation switch is closed and the second and third isolation switches are open, and the third disconnecting means are open, and control the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second and third isolation switches are closed, and the third disconnecting means are open or closed.
- 12.** An arc furnace facility comprising: an arc furnace including at least one electrode and a power supply system according to claim 11, wherein the at least one electrode is connected to at least one phase of the secondary circuit of the polyphase transformer.
- 13.** The arc furnace facility according to claim 12, wherein the arc furnace comprises a plurality of electrodes, each electrode being connected to a different phase of the secondary circuit.
- 14.** The arc furnace facility according to claim 12, wherein the power supply system further comprises a rectifier connected to the electrode and to each phase of the secondary circuit to supply the electrode with a continuous voltage from the secondary circuit.
- 15.** A method for controlling a power supply system of an arc furnace, the power supply system including: a power converter comprising an input connected to a polyphase supply grid and an output, and a polyphase transformer comprising a primary circuit and a secondary circuit, a first disconnecting means, and second disconnecting means, the method comprises: controlling the power converter to supply the electrode and to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are open, the second disconnecting means are closed, a first isolation switch is open and a second isolation switch is closed or the first isolation switch is closed and the second isolation switch is open, and third disconnecting means are open, and controlling the power converter to stabilize the current and the voltage delivered by the grid to reduce reactions in the grid when the first disconnecting means are closed, the second disconnecting means are open, the first isolation switch is open and the second isolation switch is closed, and the third disconnecting means are open or closed, the first disconnecting means connecting the input of the power converter to a first end of the first isolation switch, the second disconnecting means connecting the output of the power converter to the first end of the first isolation switch, the second end of the first isolation switch being intended to be connected to at least one electrode of the arc furnace, the second isolation switch connecting the first end of the first isolation switch to the primary circuit and the secondary circuit being connected to the second end of the first isolation switch, and the third disconnecting means connecting the input of the power converter to the output of the power converter.
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